Sensorimotor Alignment Predicts Learning in Asymmetric Perception-Action Loops

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Abstract

Embodied agents perceive the world through lossy channels but act on it directly. We define sensorimotor alignment (SA), the cosine similarity between an agent's learned policy under degraded perception and the optimal action computed from true physical state, and show that SA predicts task performance across two tasks: a 4D rock-push ($r = -0.72$, $n = 245$, $p = 5 \times 10^{-9}$ for the perception $\times$ capacity interaction) and a 6D two-rock push ($r = -0.73$, $n = 45$). SA exhibits a learnability threshold (SA ≥ 0.5 yields 98% success) and transfers across physics and appearance variants (89–106% retention). Perception quality and organism capacity interact multiplicatively. Increasing capacity reduces distance to target only when perception is sufficient ($\beta_{\text{interaction}} = -0.004$, $p = 5 \times 10^{-9}$). When perception is too degraded, additional representational power does not improve performance. An oracle-free proxy, action variance across observations, predicts performance comparably ($r = -0.82$) without access to optimal actions. These results reproduce the critical-period phenomenon observed by Blakemore and Cooper [1970], where sensory pathways that fail to align with motor experience are pruned.

1 Introduction

Embodied agents face a structural asymmetry. They perceive the world through lossy channels but act on it directly. An organism observes matter through emissions, noise, and compression, yet its motor commands modify the true physical state without passing through the perceptual chain:

$$\begin{aligned} \text{Perception:} &\quad \text{true state} \rightarrow \text{emission} \rightarrow \text{medium} \rightarrow \text{encoding} \rightarrow \text{internal model} \\ \text{Action:} &\quad \text{motor command} \rightarrow \text{directly modifies true state} \end{aligned}$$

Blakemore and Cooper [1970] demonstrated that sensory pathways which fail to achieve alignment with motor experience within the critical period are pruned; neurons in kittens' visual cortex lost selectivity for orientations never experienced. This principle has been studied extensively in developmental neuroscience [Hubel and Wiesel, 1970, Huang et al., 1999] but lacks a quantitative metric that connects perception quality, agent capacity, and task success in a controlled setting where all three are independently manipulable.

We present such a metric. We define **sensorimotor alignment** (SA): the cosine similarity between an agent's learned action under degraded perception and the analytically optimal action computed from the true state. SA quantifies the degree to which degraded perception supports correct action on the physical state the agent cannot directly observe.

We evaluate SA in a controllable simulation (WorldNN) across two tasks (4D rock-push, 6D two-rock push), 7 perception conditions, 5 capacity levels, and 7 seeds per condition. The principal findings:

1. SA predicts task performance ($r = -0.72$ on 4D, $r = -0.73$ on 6D).

2. A learnability threshold exists: SA ≥ 0.5 yields 98% success.
3. Perception and capacity interact multiplicatively ($p = 5 \times 10^{-9}$).
4. SA transfers across physics and appearance variants (89–106%).
5. SA learning slope during training predicts final success ($r = -0.71$).
6. An oracle-free proxy (action variance) predicts performance comparably ($r = -0.82$).

The paper is organized as follows. §2 reviews related work. §3 describes the WorldNN simulation. §4 defines SA and presents the metric ablation. §5 reports results across both tasks, including dynamics and transfer. §6 analyzes perception failure conditions. §7 discusses implications and limitations.

2 Related Work

Asymmetric Actor-Critic [Pinto et al., 2018] trains RL agents with privileged state information during training but deploys with observations only. SA (defined in §4) serves a complementary role: rather than using privileged information as a training signal, we use it as a post-hoc diagnostic that measures the residual alignment gap between actions under degraded perception and actions under full state access.

DreamerV3 [Hafner et al., 2023] and **IRIS** [Micheli et al., 2023] learn world models and implicitly measure prediction quality in latent space. SA is computed in true state space, not latent space; this distinction matters because the action operates on true state, not on the model's latent representation.

Quasimetric RL [Wang et al., 2023] measures directional asymmetries in value function space. SA measures asymmetry in action space. It quantifies the gap between the policy's action direction under degraded perception and the optimal direction under full state observation.

Policy cosine similarity in imitation learning is superficially similar to SA. In imitation learning, the expert and learner share the same observation. SA measures alignment across the perception gap, where the optimal action is computed from true state while the policy acts on degraded observations. This asymmetry is what makes the metric diagnostic of the perception-action loop.

Information bottleneck [Tishby et al., 2000] characterizes optimal compression-prediction tradeoffs. **Active inference** [Friston, 2010] frames the perception-action loop as free energy minimization. SA operationalizes both frameworks; low SA corresponds to high free energy and to lossy compression that destroys action-relevant information. SA differs from these frameworks in that it provides a computable scalar rather than a theoretical bound.

Critical period neuroscience [Blakemore and Cooper, 1970, Rossi et al., 1999, Huang et al., 1999]: BDNF-dependent critical periods set a biological timeout on sensory-motor alignment. If alignment is not achieved within the viability window, the pathway is pruned. In our simulation, the oracle+noise(0.5) condition produces SA ≈ 0.14, permanently below the learnability threshold (§5.2).

No prior work defines cosine alignment between a degraded-perception policy and optimal-state action across a designed perception-degradation ladder with independently controllable perception quality and organism capacity.

3 The WorldNN Simulation

3.1 The asymmetric loop

WorldNN models the full asymmetric perception-action loop. The organism's observation passes through four lossy stages, while its action bypasses all of them and directly modifies the true physical state. The rock responds to the actual force applied, regardless of the organism's perceptual accuracy.

Matter. Rock-push task. True state = $[\text{rock}_x, \text{rock}_y, \text{org}_x, \text{org}_y]$. The rock has physical properties (position, inertia, contact radius) that determine its response to force. These properties are never directly observable.

Perception chain. Matter emits 8D signals (a fixed projection of true state, analogous to light patterns). A channel adds Gaussian noise (atmospheric distortion). A VAE environment compresses the signal (sensory apparatus limitations). The organism receives the final compressed representation.

Action. The organism outputs a 2D motor command that acts directly on the matter's true state, not on the perceived state and not through the VAE in reverse. The action channel is clean.

3.2 Independently controlled variables

Table 1: Experimental variables and their ranges.

Variable	Range	Controls
Perception mode	Oracle, oracle+noise, raw emission, VAE μ	Percept degradation
env_latent_dim	8, 16, 32	VAE compression
channel_noise	0.01–0.5	Signal corruption
embedding_dim	2, 4, 8, 16, 32	Organism capacity

The oracle condition (direct true state observation) eliminates the perception chain entirely, providing the upper bound on what any perception mode can achieve. We treat embedding_dim as a working-memory-slot analogue rather than a literal cortical-neuron count; the relevant biological referent is the size of the internal state buffer that mediates between perception and action, not the gross volume of neural tissue.

4 Sensorimotor Alignment

4.1 Definition

After training, for each sampled true state s:

$$\text{SA} = \mathbb{E}_s \left[\cos\left(\pi_\theta(o(s)),\ a^*(s)\right)\right] \tag{1}$$

where $\pi_\theta(o(s))$ is the organism's action given its degraded observation and $a^*(s)$ is the analytically optimal action computed from the true state (move toward rock if far, push rock toward target if near). SA = 1 indicates perfect alignment; SA = 0 indicates orthogonal (random) action.

4.2 Metric ablation

We compared six candidate metrics as predictors of task performance on the at-scale dataset ($n = 245$), shown in Table 2.

We report cosine SA as the primary metric (interpretability, interaction sensitivity) and magnitude-weighted SA as a complementary predictor.

4.3 Theoretical grounding

SA operationalizes the sensory-motor alignment framework. Each perception modality provides an embedding e_i. A learned alignment operator R_i maps each embedding into a

Table 2: Metric ablation. Magnitude-weighted SA is the strongest overall predictor; plain cosine SA has the strongest interaction signal.

Metric	Overall r	Within-level r	Interaction F
mag-weighted SA ($\cos \times \|a\|$)	-0.893	-0.714	98.6
action magnitude $\|a\|$	-0.739	-0.582	39.4
$\lvert$SA$\rvert$ (abs cosine)	-0.727	-0.588	101.7
SA (cosine)	-0.724	-0.582	**104.8**
positive fraction (SA > 0)	-0.699	-0.568	108.5
embedding utilization	-0.499	-0.573	12.2

shared representation, from which a motor projection W_m produces actions: $a = W_m R(e)$. SA measures the quality of R, i.e., how well the learned alignment preserves action-relevant information.

5 Results

All results from the at-scale experiment: 245 trained configs + 35 random baselines. 7 perception conditions × 5 embedding dimensions × 7 seeds per condition. Random baseline (untrained organism): dist $= 0.516 \pm 0.003$, SA $= 0.003 \pm 0.022$. Success defined as dist < 0.511 (baseline -2σ).

5.1 SA predicts task performance across 7 perception conditions

Table 3: SA and task performance by perception condition (4D rock-push).

Perception mode	Mean dist ↓	Mean SA ↑	Success rate
Oracle (direct state)	0.480 ± 0.050	0.483 ± 0.142	69% (24/35)
Oracle + noise σ=0.1	0.488 ± 0.042	0.452 ± 0.138	63% (22/35)
Raw emission (8D)	0.466 ± 0.039	0.489 ± 0.090	89% (31/35)
VAE μ lat=32	0.498 ± 0.022	0.452 ± 0.066	63% (22/35)
VAE μ lat=16	0.499 ± 0.021	0.440 ± 0.076	57% (20/35)
VAE μ lat=8	0.508 ± 0.014	0.286 ± 0.057	37% (13/35)
Oracle + noise σ=0.5	0.525 ± 0.004	0.141 ± 0.077	0% (0/35)

Overall Pearson correlation: $r = -0.724$ ($n = 245$, $p < 10^{-40}$). Between the 7 condition means, $r = -0.878$. Within individual conditions, mean $r = -0.582$ (range: -0.801 to $+0.463$). Six of seven conditions show strong within-level correlation ($r < -0.7$); oracle+noise(0.5) is the exception (see §6). Figure 4 summarizes these results.

5.2 Learnability threshold

5.3 Formal interaction test: capacity × perception

A log-linear regression model predicts rock-target distance from perception quality (ordinal, 0–6), $\log_2$(emb), and their interaction:

$$\text{dist} = \beta_0 + \beta_p \cdot \text{perc} + \beta_c \cdot \log_2(\text{emb}) + \beta_{pc} \cdot \text{perc} \times \log_2(\text{emb}) + \epsilon \tag{2}$$

Full model $R^2 = 0.458$. The interaction term is highly significant: $F(1, 241) = 34.2$, $p = 5.0 \times 10^{-9}$. The negative interaction coefficient ($\beta_{pc} = -0.004$) confirms that capacity and perception are multiplicative, not additive.

Table 4: Success rate by SA threshold.

SA threshold	N above	Success rate
≥ 0.3	187	68%
≥ 0.4	124	**88%**
≥ 0.5	53	**98%**
≥ 0.6	18	**100%**
< 0.3	58	7%

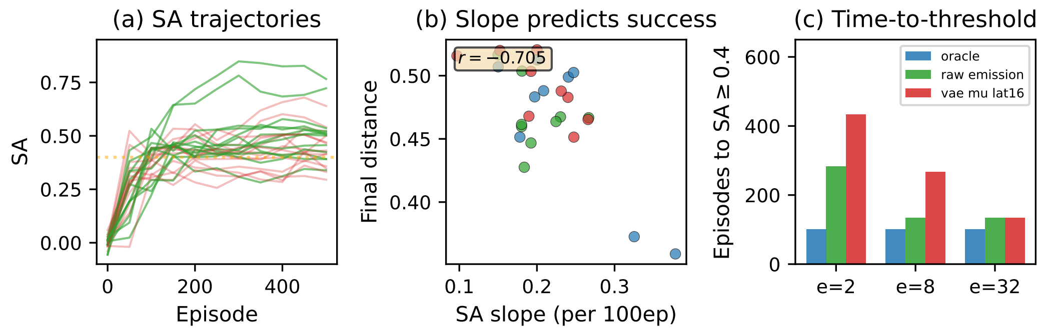

Figure 1: SA dynamics during training. (A) SA trajectories for 27 configs over 500 episodes; trajectories diverge by episode 200. (B) SA slope over the first 200 episodes vs. final rock-target distance ($r = -0.705$, $n = 27$). (C) Time-to-threshold (episode at which SA exceeds 0.4) vs. final distance.

5.4 SA dynamics during training

The slope of SA over the first 200 episodes correlates with final rock-target distance at $r = -0.705$, stronger than any single snapshot (e.g., SA at episode 100: $r = -0.232$). The rate of alignment improvement is more predictive than the alignment level at any given time (Figure 1).

5.5 Second task: 2-rock push (6D state)

We implemented a 2-rock variant with 6D state, 12D emission, and 2D action. The SA-performance correlation at 6D is $r = -0.728$ ($n = 45$), matching the 4D result ($r = -0.724$). The capacity gradient replicates across all perception modes: oracle SA rises from 0.41 (emb=4) to 0.71 (emb=64).

We also tested a 3-rock variant (8D state, 105 configs). The correlation is weaker ($r = -0.300$, $p = 4 \times 10^{-4}$) due to a floor effect; the perception $\times$ capacity interaction remains significant ($F(1, 101) = 12.5$, $p = 4 \times 10^{-4}$).

5.6 SA transfer across physics and appearance variants

SA transfers across both physics variants (push radius ±20%, push strength ±30%; 94–106% retention) and appearance variants (emission matrix perturbed by $\epsilon \times \mathcal{N}(0, 1)$, $\epsilon \in \{0.1, 0.3, 0.5\}$; $r(\epsilon, \text{retention}) = 0.033$). Appearance perturbation has no measurable effect on SA retention.

5.7 Information-theoretic bound on the SA ceiling

The SA ceiling achievable by any organism is bounded by the mutual information between the environment state and the organism's sensory channels. We tested this directly by

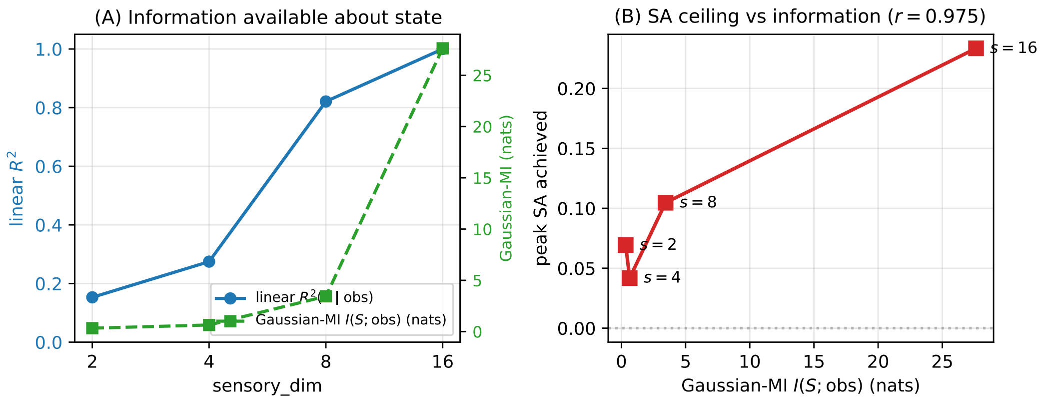

Figure 2: Information-theoretic bound on the SA ceiling. (A) Linear-probe $R^2(S \mid \text{obs})$ (left axis) and Gaussian-MI lower bound on $I(S; \text{obs})$ (right axis) as a function of sensory_dim. (B) Peak SA achievable at each sensory_dim plotted against Gaussian-MI; Pearson $r = 0.975$.

independently sweeping sensory richness and model capacity.

In a sensory-capacity grid with emission_dim$= 16$, sensory_dim $\in \{2, 4, 8, 16\}$, and embedding_dim $\in \{2, 4, 8, 16, 32\}$ (100 configs, 5 seeds each, 800 episodes per config), peak SA rose monotonically with sensory richness: 0.069, 0.042, 0.105, 0.234 for sensory_dim $= 2, 4, 8, 16$. A floor effect was evident at sensory_dim ≤ 4: no embedding dimension rescued performance. Conversely, rich input with minimal capacity (sensory$= 16$, embed$= 2$, SA$= 0.033$) outperformed poor input with maximal capacity (sensory$= 2$, embed$= 32$, SA$= -0.007$); Cohen's $d = 1.10$, 95% bootstrap CI for mean difference $[+0.004, +0.083]$.

Estimating the information available about state S from the first sensory_dim emission channels via a linear-probe Gaussian-MI lower bound, we obtain $I(S; \text{obs}) \approx \{0.33, 0.64, 3.44, 27.6\}$ nats for sensory_dim $= \{2, 4, 8, 16\}$, with linear $R^2(S \mid \text{obs}) = \{0.15, 0.28, 0.82, 1.00\}$. The Gaussian-MI correlates with peak SA at $r = 0.975$ (Figure 2). This matches the Data Processing Inequality prediction: capacity cannot recover information absent from the input. The floor effect at sensory ≤ 4 corresponds to linear $R^2 < 0.3$. The information about state is below what the task requires, and no policy dimension closes the gap.

The 2–16 channel range sits at the low end of biological sensing, comparable to paramecium chemotaxis (2), vestibular (~ 6), haptic vibrotactile (~ 10), and cutaneous types per digit (~ 16); well below vision ($\sim 10^6$ optic nerve fibers), audition ($\sim 10^4$ cochlear hair cells), or olfaction (10^2–10^3 receptor classes).

Scope on 2-rock. A direct replication on the 2-rock task (60 configs, 3 seeds, 800 episodes) produced a floor effect: overall mean distance 0.501 (near-random), peak SA 0.098 vs. 0.234 on 1-rock, and no monotonic sensory-richness ceiling. This matches the 3-rock floor observed previously: PPO training budget is insufficient when reward is diluted across multiple targets. The 2-rock result does not contradict the 1-rock finding, but restricts the empirical support for the information bound to tasks that converge within the training budget.

6 Perception Failure Conditions

Three conditions produce SA values below the learnability threshold, analogous to the pruned sensory pathways observed by Blakemore and Cooper [1970].

Stochastic VAE (z). Sampling noise destroys the spatial signal. SA ≈ 0 across all conditions and all capacity levels.

VAE lat=8 (μ). Extreme compression caps SA at approximately 0.29.

Oracle + noise σ=0.5. SA $= 0.14 \pm 0.08$, below the learnability threshold. All 35 configs fail (0% success). This condition shows a within-level correlation reversal ($r = +0.46$), attributable to near-zero distance variance (std = 0.004): when all configs fail equally, any correlation with SA reflects noise. VAE lat=8, with comparable SA (0.29 ± 0.06), shows normal negative correlation ($r = -0.75$) because it has sufficient distance variance (std = 0.015).

7 Discussion

7.1 The perception-action asymmetry

The asymmetry between lossy perception and direct action is a structural property of any embodied system. SA quantifies a specific consequence: whether the information surviving the perception chain is sufficient to determine correct actions on a physical state the agent cannot directly observe. The combination of a defined metric and a controllable testbed where every stage of the loop is independently manipulable is the methodological contribution.

7.2 Relationship to biological critical periods

The perception failure conditions in §6 reproduce the computational structure of critical-period pruning. In biological systems, BDNF-dependent signaling sustains sensory pathways that achieve alignment with motor experience; pathways that do not are eliminated [Huang et al., 1999, Rossi et al., 1999]. In our simulation, oracle+noise(0.5) produces SA permanently below the learnability threshold. The computational analog of pruning is the flat SA landscape: no learning signal exists to improve alignment. The analogy is structural, not mechanistic: we do not model BDNF or synaptic pruning.

A testable biological prediction follows: in a behavioral animal model, monocular deprivation during the BDNF-dependent critical period should drive SA in visuomotor reaching tasks below the learnability threshold (analogous to our oracle+noise(0.5) regime), and the eventual deprived-eye reaching deficit should correlate with the integrated below-threshold SA duration rather than with the absolute SA endpoint. The pruning theory predicts a critical-period cutoff; the SA-as-diagnostic framing predicts that the integrated time-below-threshold is the right dose-response variable.

7.3 SA as a training diagnostic

SA slope during the first 200 episodes predicts final performance ($r = -0.705$). All configurations with SA slope < 0.1 per 100 episodes at episode 200 failed to learn. This suggests SA slope as an early stopping criterion, though cross-task validation is needed.

7.4 Transfer and structural alignment

SA transfers across physics variants (94–106% retention) and appearance variants ($r(\epsilon, \text{retention}) = 0.033$). The organism retains alignment when evaluated on rocks that both look and respond differently, indicating that SA captures the structure of the push interaction rather than a memorized mapping.

7.5 Oracle-free proxy estimation

SA as defined requires oracle access to the optimal action $\mathbf{a}^*$, limiting its applicability to settings where the optimal policy is computable. We evaluated five oracle-free proxy candidates across 30 configurations (2 perception conditions $\times$ 3 embedding dimensions $\times$ 5 seeds). The strongest proxy is *action variance*: $\text{Var}[\mathbf{a}(o)]$ across randomly sampled

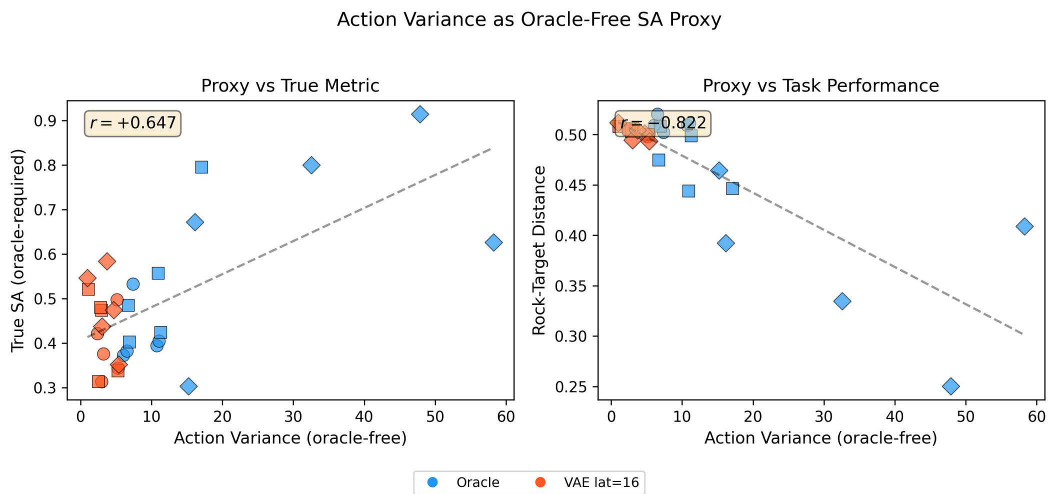

Figure 3: Action variance as an oracle-free SA proxy. (a) Action variance vs. true SA ($r = +0.65$). (b) Action variance vs. task performance ($r = -0.82$), comparable to true SA's $r = -0.80$. Oracle-free, computable from policy outputs alone.

observations o. Action variance measures whether the organism produces structured, state-dependent actions (high variance) or outputs similar actions regardless of observation (low variance, indicating no learned mapping).

Action variance predicts task performance with $r = -0.82$ ($n = 30$), comparable to true SA ($r = -0.80$ on the same grid). Within oracle-only conditions, $r = -0.78$; within VAE-only conditions, $r = -0.75$. Four other candidates, namely prediction consistency (forward model error), action stability under noise, value-action alignment, and policy self-predictability, achieved weaker correlations ($|r| \leq 0.43$). Action variance requires no knowledge of the optimal action, the environment dynamics, or the true state, and is computable from the organism's policy network alone (Figure 3).

7.6 Limitations

The 4D and 6D tasks are low-dimensional. The 3-rock (8D) task produced weaker SA correlation ($r = -0.300$) due to a floor effect from insufficient training budget. The learnability threshold (SA $\approx$ 0.4–0.5) may be task-specific. The oracle+noise(0.5) reversal ($r = +0.46$) indicates SA is unreliable when perception carries zero task-relevant information. While action variance provides a strong oracle-free proxy on these tasks, cross-task validation is needed to confirm generalization.

Noise model sensitivity. Recomputing the recon ceiling under signal-dependent noise ($\sigma_i \propto |y_i|$, Weber-style) gives the same curve as Gaussian within ± 0.05 R^2 when matched on effective per-channel σ; the information-theoretic argument is invariant to the biological noise-model choice.

SA versus reconstruction-loss. We computed the linear-probe recon ceiling $R^2(\text{state} \mid \text{obs})$ for each of the seven perception conditions in the at-scale grid (§5, $n = 245$): oracle 1.00, oracle+noise(0.1) 0.85, oracle+noise(0.5) 0.19, raw emission 0.96, VAE μ lat 8 0.45, lat 16 0.82, lat 32 0.86. Per-config recon ceilings correlate with task distance at $r = -0.436$, weaker than SA's $r = -0.724$. After controlling for recon ceiling, SA still correlates with distance at partial $r = -0.679$; recon's residual partial correlation flips sign to $+0.285$, indicating shared variance with SA rather than independent predictive power. Multiple regression on

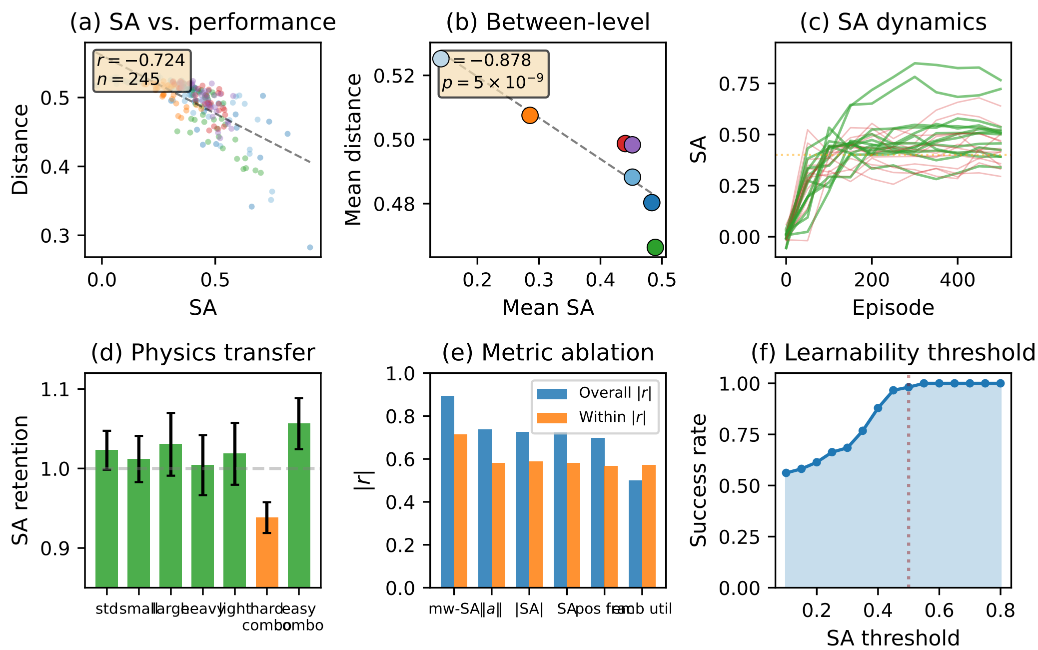

Figure 4: Sensorimotor alignment predicts learning across perception and capacity conditions. (a) SA vs. distance ($r = -0.724$, $n = 245$). (b) Between-level correlation ($r = -0.878$, interaction $p = 5 \times 10^{-9}$). (c) SA trajectories during training; green = success, red = failure. (d) SA retention across physics variants (94–106%). (e) Metric ablation: 6 candidates compared. (f) Learnability threshold curve.

distance gives $R^2 = 0.190$ for recon alone and $R^2 = 0.563$ when SA is added: SA contributes $\Delta R^2 = +0.374$ on top of reconstruction. SA captures a structural property of the policy that reconstruction loss does not.

This finding is regime-dependent. On a compressed grid where every perception condition saturates the same recon ceiling — as in the §5.7 sensory-capacity sweep, where sensory_dim ≤ 4 fails to learn regardless of capacity — SA's incremental contribution drops to $\Delta R^2 = 0.004$. SA is informative when perception itself varies; on grids where it does not, recon and SA collapse to the same signal. The asymmetric perception-action loop is where SA pulls its weight.

8 Conclusion

We defined sensorimotor alignment (SA), a scalar metric that quantifies how well an agent's learned policy under degraded perception aligns with the optimal action on true physical state. Across 245 conditions on a 4D task and 45 on a 6D task, SA predicts performance ($r = -0.72$ and $r = -0.73$), exhibits a learnability threshold (SA ≥ 0.5 yields 98% success), and reveals a multiplicative perception $\times$ capacity interaction ($p = 5 \times 10^{-9}$). SA transfers across physics and appearance variants (89–106% retention), its training slope predicts final success ($r = -0.71$), and action variance provides a comparable oracle-free proxy ($r = -0.82$). These results formalize the principle that sensory pathways failing to achieve motor alignment are functionally useless, reproducing the computational structure of critical-period pruning in a controlled simulation.

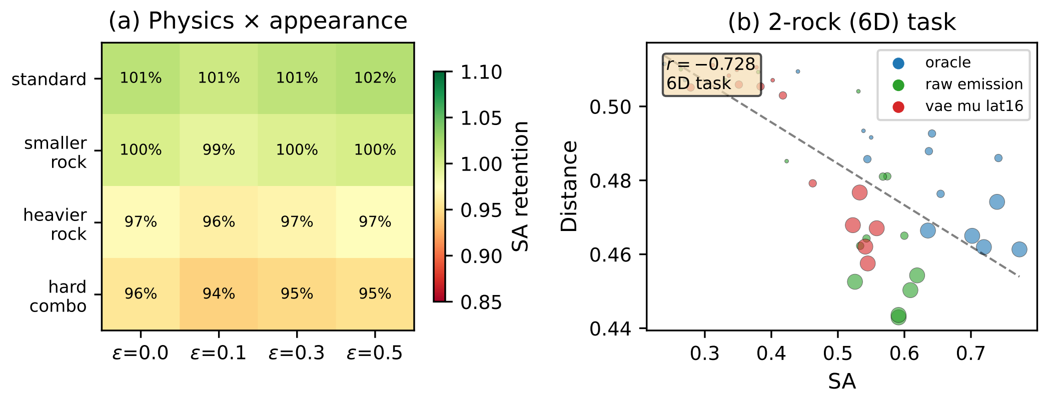

Figure 5: SA generalizes across tasks and variants. (a) SA retention under combined physics and appearance perturbation ($r(\epsilon, \text{retention}) = 0.033$). (b) 2-rock task (6D state): SA-performance correlation $r = -0.728$ replicates the 4D finding.

Supplementary Materials

Code, data, and all experimental scripts are available at `https://github.com/dfu99/WorldNN`. The repository includes: the WorldNN simulation framework (`src/worldnn/`), all experiment scripts (`experiments/`), and raw results in JSON format (`results/`).

Additional figures (SA dynamics detail, per-objective plots) are provided in the `results/` directory.

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